Fiscal Year:	FY 2021	**Task Last Updated:**	FY 03/30/2021
PI Name:	Perera, Imara Y Ph.D.		
Project Title:	Transcriptional and Post Transcriptional Regulation of Seedling Development in Microgravity		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	None		
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Zip Code:	27695	**Congressional District:**	4
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	11/01/2014	**End Date:**	12/31/2020
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
Contact Email:	Yuri.V.Griko@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: Extended to 12/31/2020 per NSSC information (Ed., 8/3/2020) NOTE: Extended to 12/31/2019 per F. Hernandez/ARC (Ed., 5/6/19) NOTE: Extended to 4/30/2019 per F. Hernandez/ARC (Ed., 11/2/17)		
Key Personnel Changes/Previous PI:	August 2017: Graduate Research Assistant Eric Land was the technician on the project who carried out flight build and will be involved in all post flight processing.		
COI Name (Institution):			
Grant/Contract No.:	NNX15AB07G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Plants are a vital part of human life support systems for long-duration spaceflight and habitation. However, the space environment is not optimal for plant growth. Plants grown in space are subject to many unfamiliar stresses (in addition to the lack of gravity) and recent transcriptional profiling studies indicate that there are global changes in gene expression between space and ground controls. Post transcriptional regulation of RNA is emerging as an important mechanism of modulating gene expression under different environmental conditions. To date however, there have been no studies to examine the role of small regulatory RNAs in plant responses to the space environment. We propose to examine the transcriptional and post transcriptional mechanisms that regulate early seedling development in space and microgravity. Our hypothesis is that plant adaptation and response to the space environment will involve novel regulatory small RNAs. Our previous flight experiment has revealed novel regulatory mechanisms and provides the foundation for further investigation and the proposed research. The long term goals of this research are to understand the molecular mechanisms by which plants sense and adapt to changes in their environment and to characterize the regulatory networks that mediate these responses. This knowledge will be valuable for designing plants that are better able to withstand spaceflight, microgravity, and adverse environmental conditions. This project is in alignment with P2, one of the highest priority recommendations of the Space Biology Research focus on Plant and Microbial Biology as outlined in the Decadal Survey Report, “to analyze plant growth and physiological responses to the multiple stimuli encountered in space flight environments.”
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will have relevance and Earth benefits on many levels. On a practical level, we will optimize and develop protocols for efficient handling of limited experimental material that has application to many ground based studies. Most importantly, the knowledge gained from this work will provide a framework for designing and improving plants that are better adapted to adverse environments, which has potential benefit on Earth in the face of global climate change. In addition, this work will contribute towards enhancing education by providing ‘hands on’ training to undergraduate students in Science-Technology-Engineering-Math (STEM) programs.
Task Progress:	FINAL REPORTING -- MARCH 2021 Although the uneven growth and germination of seedlings during the Plant RNA Regulation (PRR) spaceflight limited our ability to accomplish our originally proposed objectives, we have nevertheless been successful in carrying out transcriptional profiling of both root and shoot samples. In addition, we were able to also profile sRNAs in shoot tissue and we identified 3 miRNAs that appear to specifically regulate genes encoding proteins with chloroplastic functions. Transcriptional profiling and comparison with our previous spaceflight experiment has revealed consistent patterns of regulation between the two experiments. As reported by other groups, we detected cell wall associated genes to be downregulated in micro g, probably due to the reduced mechanical load. Our results also identify the chloroplast (and chloroplast related functions) as being highly sensitive to the spaceflight environment. Surprisingly, our work has uncovered significant downregulation in micro g of genes involved in stress responses, a result which is counter to some published reports, which compared spaceflight to ground controls. This highlights the importance of having a “1g” control in space, which enabled us to identify primary effects of micro g from overall spaceflight related responses. ANNUAL REPORTING AS OF SEPTEMBER 2020 1. The Principal Investigator (PI) and graduate student attended ASGSR (American Society for Gravitational & Space Research) in November 2019 in Denver CO. The PI was selected to present a short talk describing this work entitled “Conserved plant transcriptional responses to microgravity from two consecutive spaceflight experiments.” 2. Given the success with the shoot samples, we went ahead and processed all of the root samples. We isolated RNA for root samples representing the 5 different fractional g levels within the ‘1g” rotor as well as micro g from day 4 and day 6 and day 6 ground controls. 3. The RNA samples were submitted for library preparation and NextGen Sequencing to be carried out “in house” at the Genomics Science Laboratory at North Carolina State University (NCSU) in December 2019. Due to their back log, the facility had started the quality check on the samples in March 2020 right before research activities were severely impacted due to the virus pandemic. 4. During the period of March to July, we were under reduced mandatory personnel only operation and therefore were unable to carry out much lab work. During this time the PI worked on data compilation and analysis. 5. Since July 2020 when normal operations were resumed at the Genomic Sciences Laboratory (GSL) we have just received the raw data for the root samples. We are currently carrying out the initial data processing and analysis. We anticipate that this work will be completed within the next few months.
Bibliography Type:	Description: (Last Updated: 05/30/2025)
Articles in Peer-reviewed Journals	Land ES, Sheppard J, Doherty CJ, Perera IY. "Conserved plant transcriptional responses to microgravity from two consecutive spaceflight experiments." Front Plant Sci. 2024 Jan 8;14:1308713. https://doi.org/10.3389/fpls.2023.1308713 ; PMID: 38259952; PMCID: PMC10800490 , Jan-2024